

CHUNCHEON, Korea, Republic of

WMO: 471010

Lat: 37.9025N

Lon: 127.7358E

Elev: 79

StdP: 100.38

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.8	-12.7	-22.2	0.5	-9.1	-20.0	0.6	-8.3	6.5	-2.4	5.4	-3.1	0.7	0	0.379

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	32.9	24.7	31.5	23.9	30.2	23.1	26.5	30.2	25.7	29.2	25.0	28.2	2.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.6	21.1	28.1	24.8	20.0	27.4	24.1	19.1	26.9	83.5	30.2	80.1	29.4	76.9	28.2	32.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.3	4.4	3.8	-17.7	34.8	2.8	1.2	-19.7	35.6	-21.4	36.3	-23.0	37.0	-25.0	37.8		
(5)			-18.2	27.8	2.7	1.4	-20.1	28.8	-21.7	29.6	-23.3	30.4	-25.2	31.5	(4)	(5)

Monthly Climatic Design Conditions

			Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
			(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	-3.9	-0.7	5.1	11.8	17.7	22.4	25.3	25.5	20.4	13.8	5.8	-1.7	(6)
(7)		DBStd	10.80	4.09	3.84	3.56	3.72	2.75	2.33	2.29	2.52	2.71	3.51	4.19	4.12	(7)
(8)		HDD10.0	1417	430	299	156	23	0	0	0	0	11	136	362		(8)
(9)		HDD18.3	3023	688	533	409	196	44	2	0	11	144	376	621		(9)
(10)		CDD10.0	2103	0	0	5	78	240	373	474	482	313	128	11	0	(10)
(11)		CDD18.3	669	0	0	0	2	26	125	216	224	74	3	0	0	(11)
(12)		CDH23.3	5304	0	0	0	43	301	952	1736	1881	382	11	0	0	(12)
(13)		CDH26.7	1664	0	0	0	4	56	293	577	672	62	0	0	0	(13)
(14)	Wind	WSAvg	1.2	1.1	1.2	1.6	1.6	1.4	1.3	1.2	1.2	1.1	1.0	1.1	1.1	(14)
(15)	Precipitation	PrecAvg	1324	19	26	38	75	96	130	381	298	144	49	46	22	(15)
(16)		PrecMax	2069	82	96	109	220	282	460	933	785	492	137	142	79	(16)
(17)		PrecMin	675	0	0	6	2	12	21	117	33	5	2	7	2	(17)
(18)		PrecStd	302	17	22	25	50	56	97	214	172	121	34	32	16	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.5	12.7	20.0	27.1	30.3	32.9	34.5	34.7	30.1	24.7	18.6	9.8	(19)
(20)			MCWB	3.7	5.9	11.0	15.3	17.6	20.9	25.9	25.7	22.6	17.1	12.4	5.8	(20)
(21)		2%	DB	5.5	10.0	17.1	24.4	28.2	31.1	32.8	33.1	28.4	22.8	16.2	7.4	(21)
(22)			MCWB	1.8	4.6	9.2	14.1	16.9	20.6	25.0	25.4	21.2	16.5	11.3	4.1	(22)
(23)		5%	DB	3.8	7.8	14.7	22.1	26.7	29.8	31.3	31.8	27.1	21.0	14.3	5.9	(23)
(24)			MCWB	0.6	3.1	7.7	12.8	16.5	20.2	24.3	25.0	20.3	15.4	10.3	3.1	(24)
(25)		10%	DB	2.2	5.7	12.5	20.1	24.9	28.4	29.8	30.4	25.6	19.5	12.4	4.3	(25)
(26)			MCWB	-0.4	1.7	6.3	11.6	15.7	19.8	23.8	24.4	19.7	14.5	9.0	1.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.4	7.2	13.0	18.1	20.4	24.1	27.6	27.9	24.5	19.7	14.3	6.6	(27)
(28)			MCDWB	6.6	10.5	17.5	23.9	25.2	28.6	31.2	31.0	28.1	22.4	16.6	8.5	(28)
(29)		2%	WB	2.4	5.2	10.3	15.8	19.0	22.8	26.4	26.8	23.1	17.8	12.5	4.7	(29)
(30)			MCDWB	4.6	9.3	15.1	21.3	24.0	26.6	30.0	30.6	26.1	20.7	15.1	6.6	(30)
(31)		5%	WB	1.2	3.5	8.6	14.1	18.0	22.0	25.6	26.1	22.0	16.6	10.9	3.5	(31)
(32)			MCDWB	2.9	6.8	13.5	19.5	23.4	26.1	29.1	29.7	24.6	19.5	13.1	5.4	(32)
(33)		10%	WB	0.1	2.1	6.9	12.7	17.2	21.3	25.0	25.4	21.0	15.4	9.6	2.1	(33)
(34)			MCDWB	1.9	5.1	11.7	18.2	22.4	25.7	28.0	28.8	24.1	18.4	11.9	4.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.2	11.1	11.7	13.2	12.2	10.3	7.2	7.9	9.4	11.0	9.7	9.3	(35)
(36)			MCDBR	10.3	13.4	15.4	17.0	15.6	13.3	9.9	9.9	10.8	11.7	11.9	9.9	(36)
(37)			MCWBR	7.0	8.3	8.4	7.7	5.8	4.4	3.5	3.5	4.3	5.7	7.0	6.8	(37)
(38)		5% WB	MCDWB	8.6	11.9	13.8	13.8	12.2	9.5	8.0	8.4	8.2	9.6	9.6	9.0	(38)
(39)			MCWBR	6.5	7.9	8.4	7.5	5.7	4.1	3.5	3.6	3.9	5.4	6.3	6.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.419	0.481	0.573	0.623	0.629	0.693	0.662	0.574	0.497	0.491	0.466	0.416	(40)	
(41)		taud	1.925	1.820	1.691	1.641	1.668	1.603	1.721	1.930	2.063	1.954	1.951	1.975	(41)	
(42)		Ebn at Noon	708	714	689	687	695	651	667	718	752	699	655	679	(42)	
(43)		Edh at Noon	140	178	223	248	246	262	231	184	152	154	137	125	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.46	3.26	4.14	4.88	5.48	5.12	3.98	4.15	3.98	3.49	2.41	2.19	(44)	
(45)		RadStd	0.18	0.26	0.43	0.40	0.51	0.45	0.58	0.57	0.47	0.26	0.27	0.18	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+120	+60	(46)
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+99	+53	(47)

Nomenclature: See separate page